Faculté de Médecine de Bejaia
Semiologie medicale
UE 1

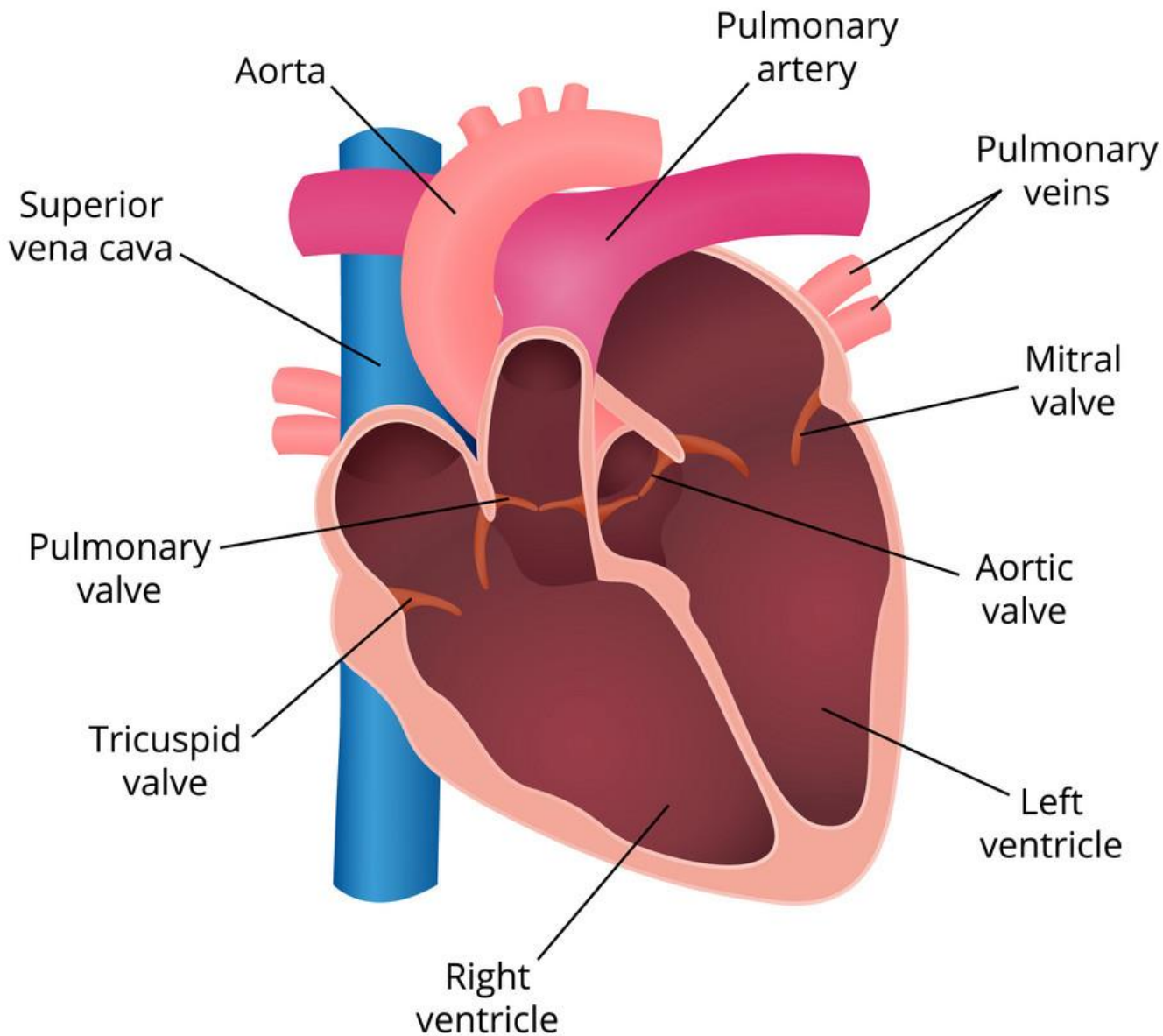
Electrocardiogramme : Généralités

Dr. Terra Dr. Bennour

Electrocardiogramme
ECG



HEART ANATOMY



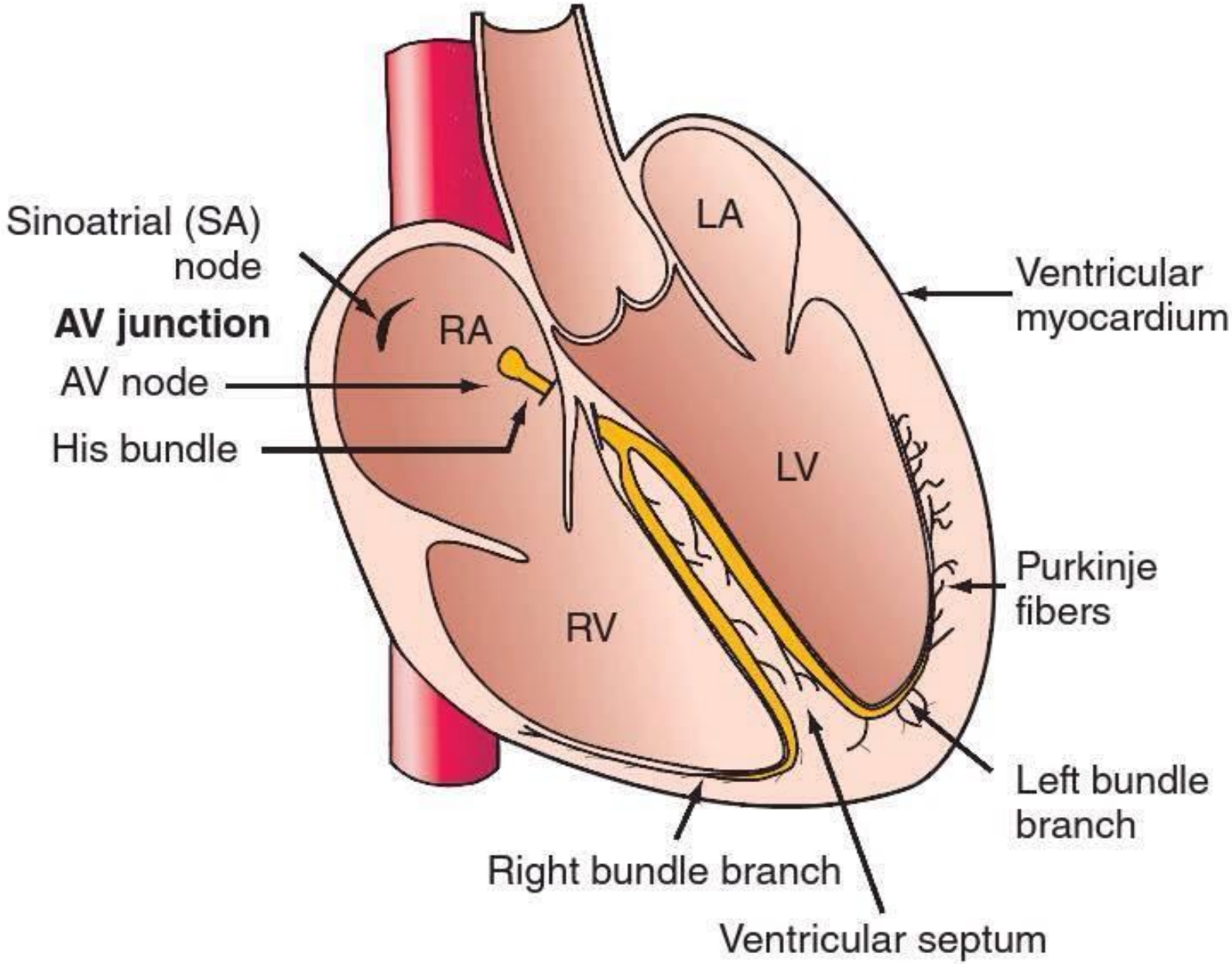
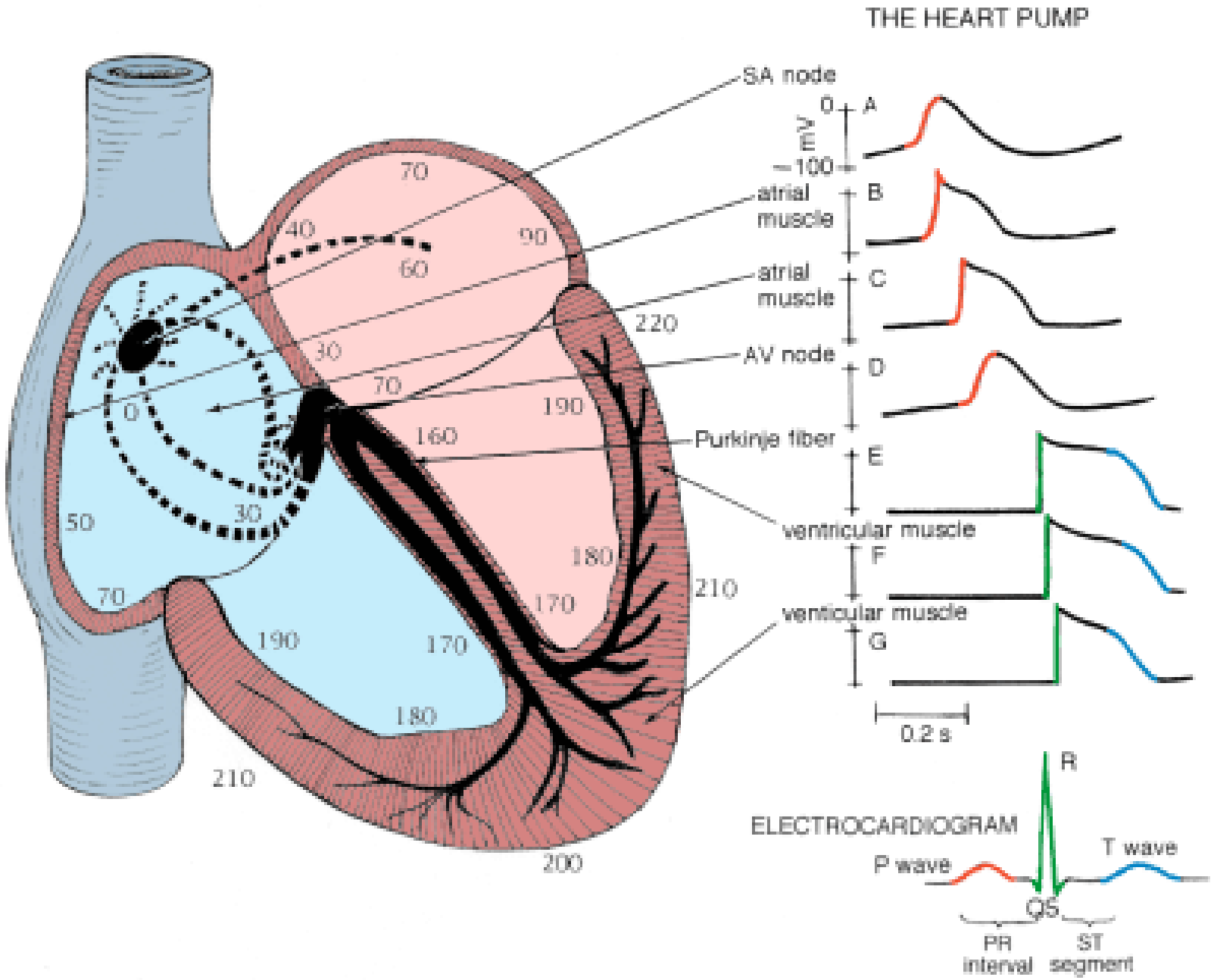


FIGURE 268-1 **Schematic of the cardiac conduction system.**


THE HEART PUMP
SA node
atrial muscle
atrial muscle
AV node
Purkinje fiber
ventricular muscle
ventricular muscle
mV
0
−100
A
B
C
D
E
F
G
0.2 s
70
40
90
60
30
220
0
70
190
160
50
30
180
70
210
170
190
170
180
210
200
R
ELECTROCARDIOGRAM
T wave
P wave
QS
PR interval
ST segment

Introduction

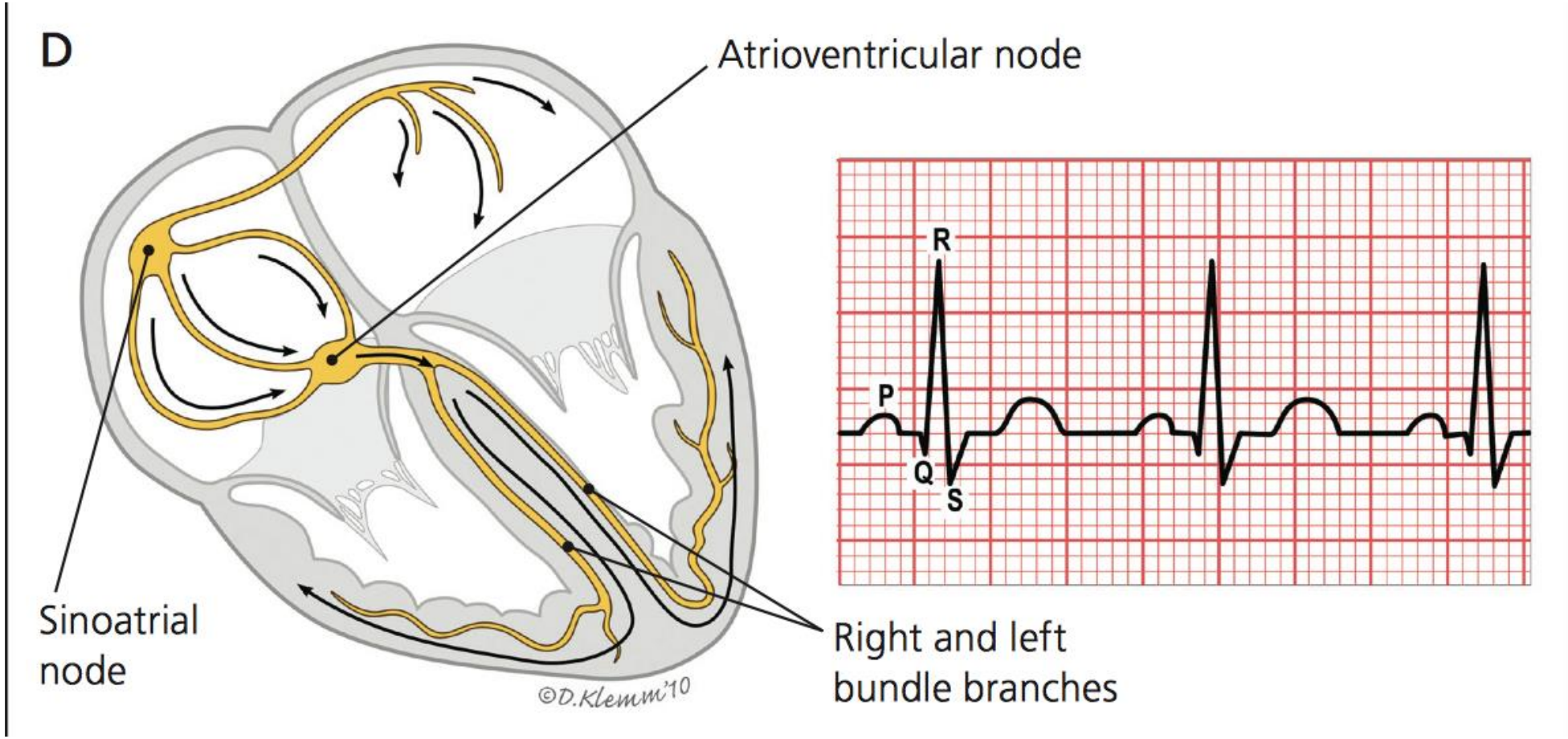
- Traduit l’activité électrique du cœur
- Ondes et segments
- Déflexions positives et négatives
- Obtenir un tracé des polarisations et des dépolarisations des oreillettes et des ventricules



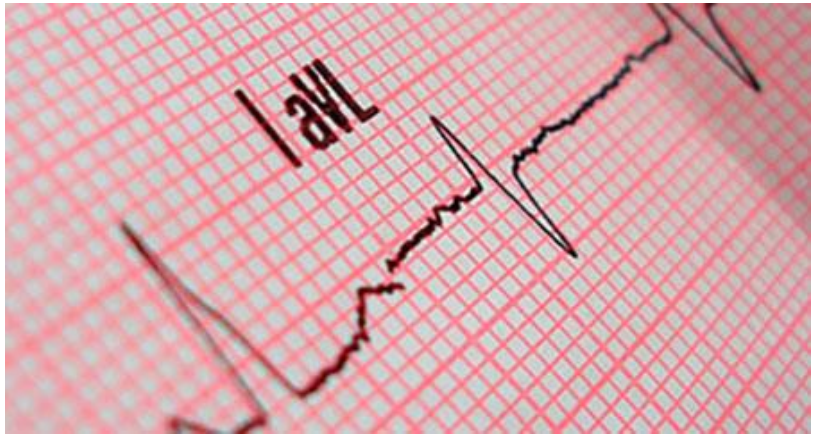
Introduction

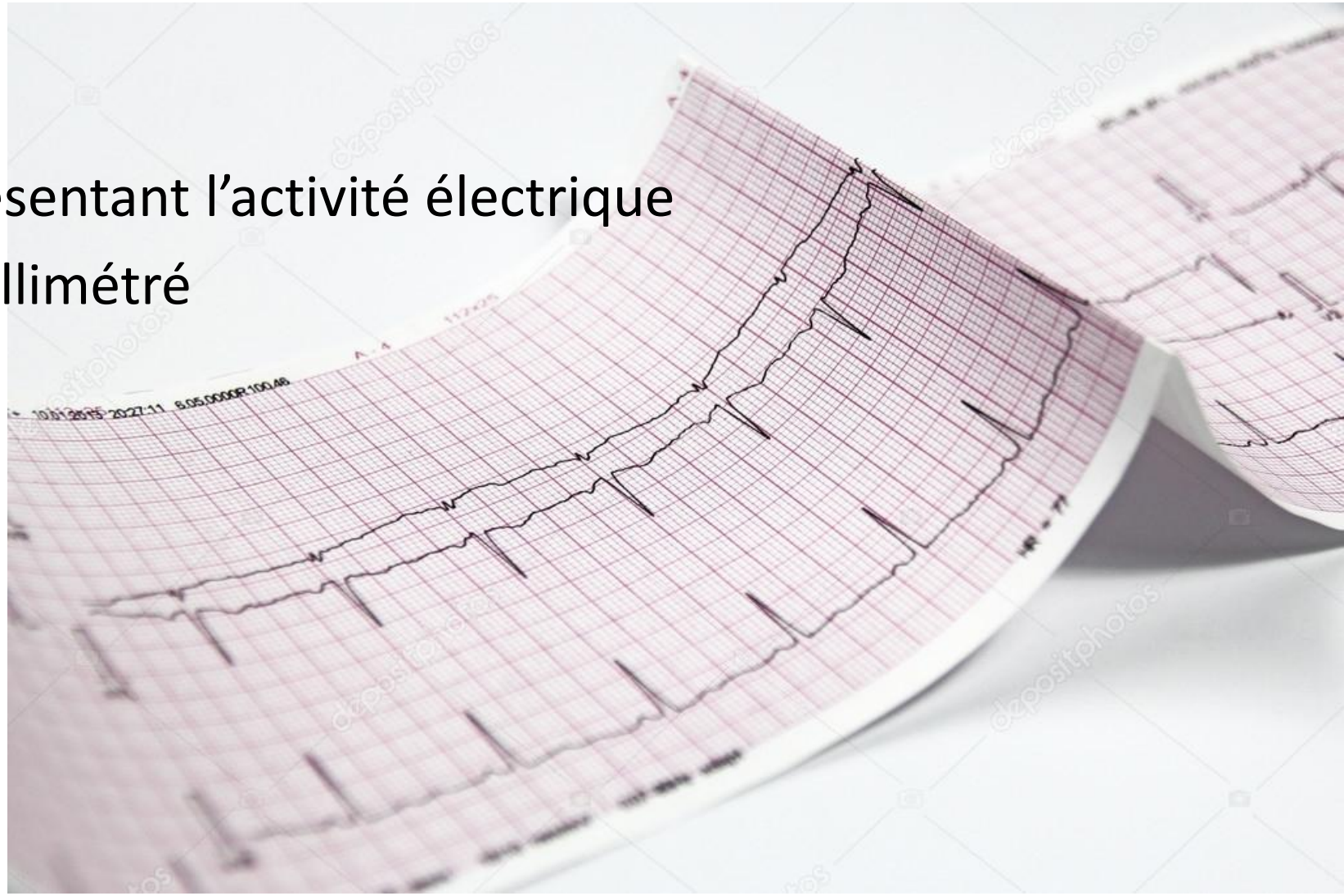
- Détecte des troubles du rythme et de la conduction
- Troubles de la conduction
- Ischémie
- Modification structurales cardiaques
- Retentissement des maladies du cœur (HTA, valvulopathies)




D
Atrioventricular node
R
P
Q
S
Sinoatrial node
Right and left bundle branches
©D.Klemm'10

- ECG = graphique représentant l'activité électrique du cœur sur papier millimétré





électrodes

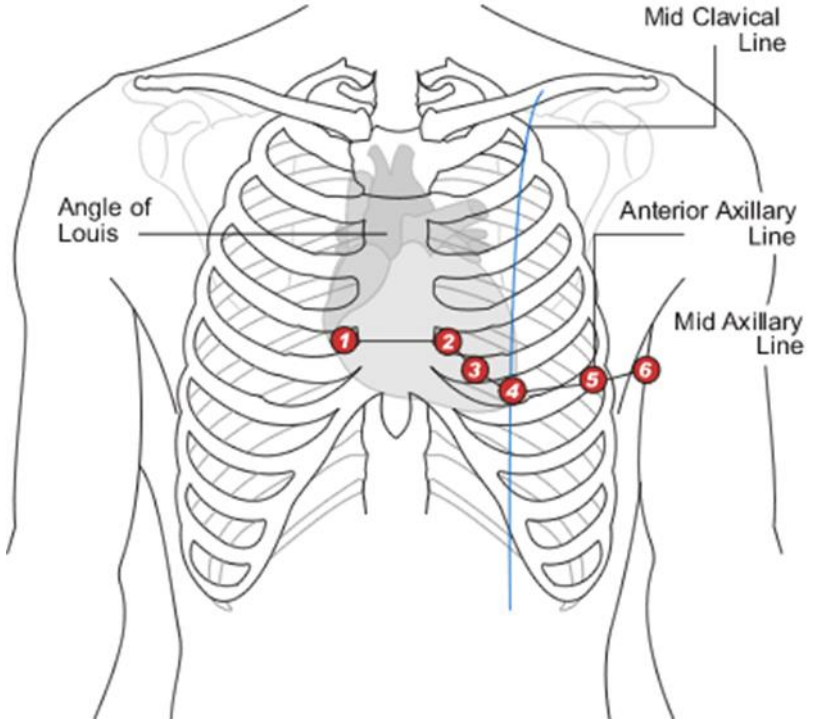


Appareil ECG





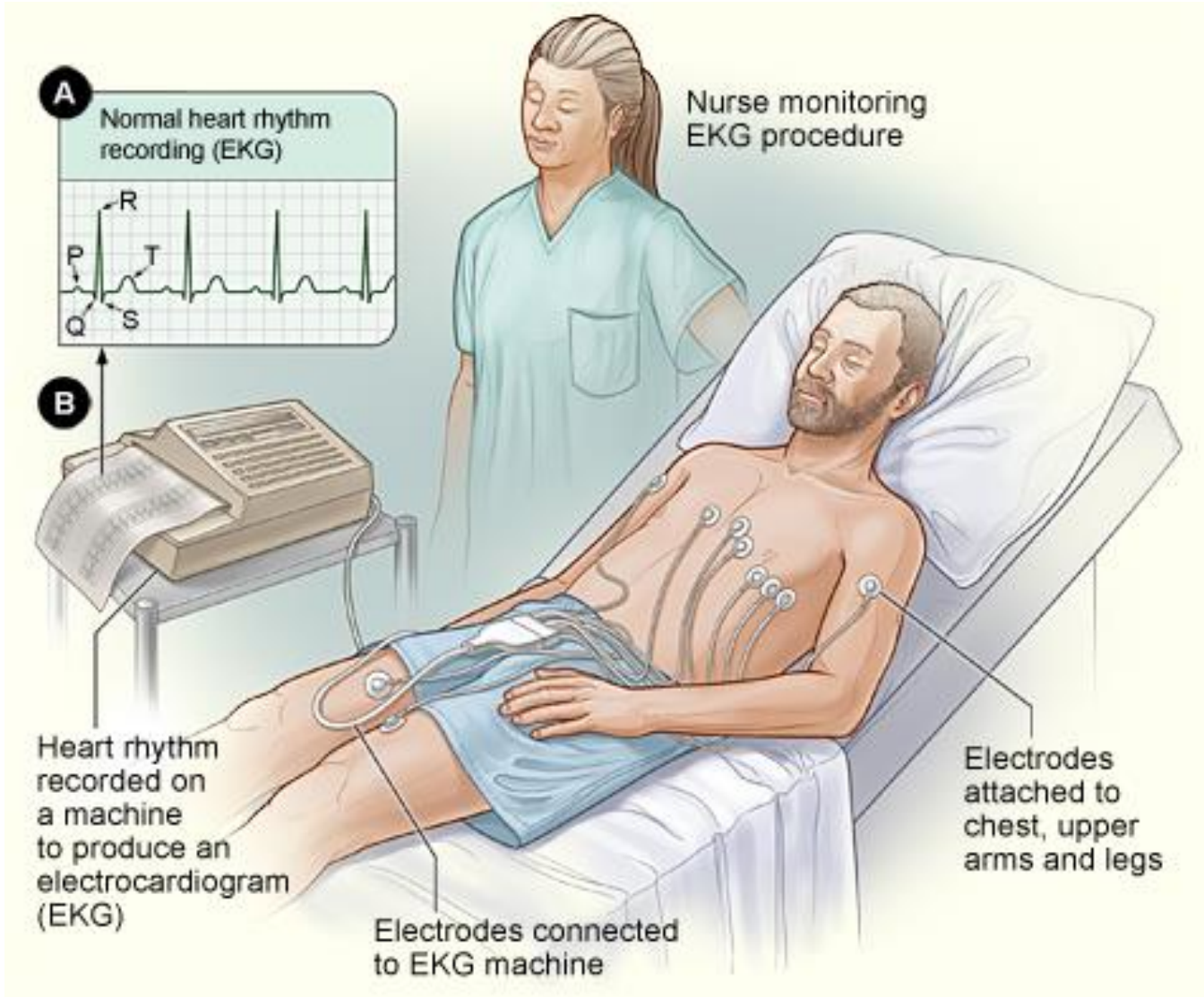
- tracé de base : 12 dérivations dont
- 6 périphériques (D1, D2, D3, VF, VR, VL)
- 6 précordiales (V1-V6)

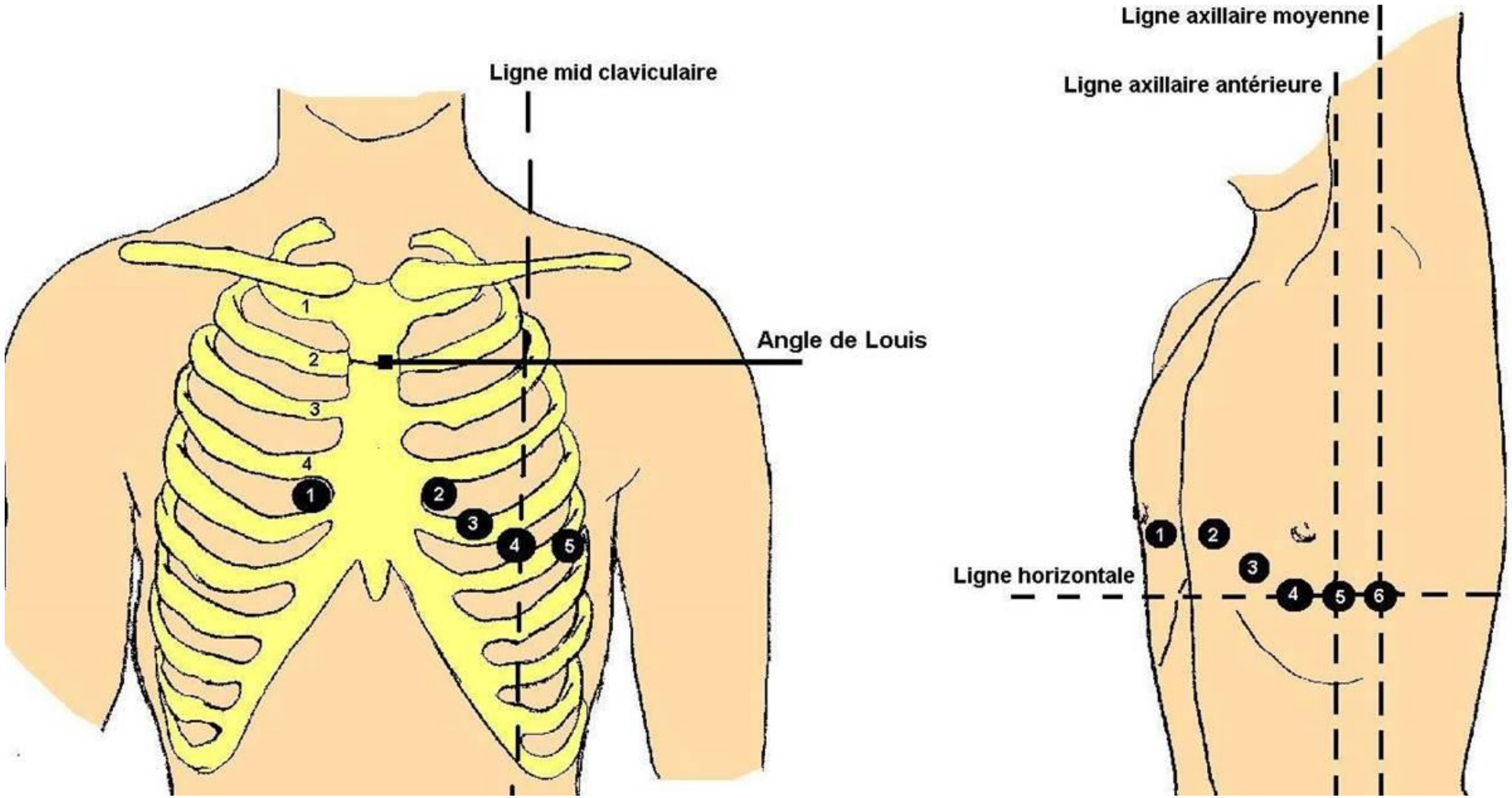


Il existe d'autres dérivations, utilisées surtout dans les syndrome coronariens

- V3R (VD)
- V4R (VD)
- V7, V8, V9 : base du cœur

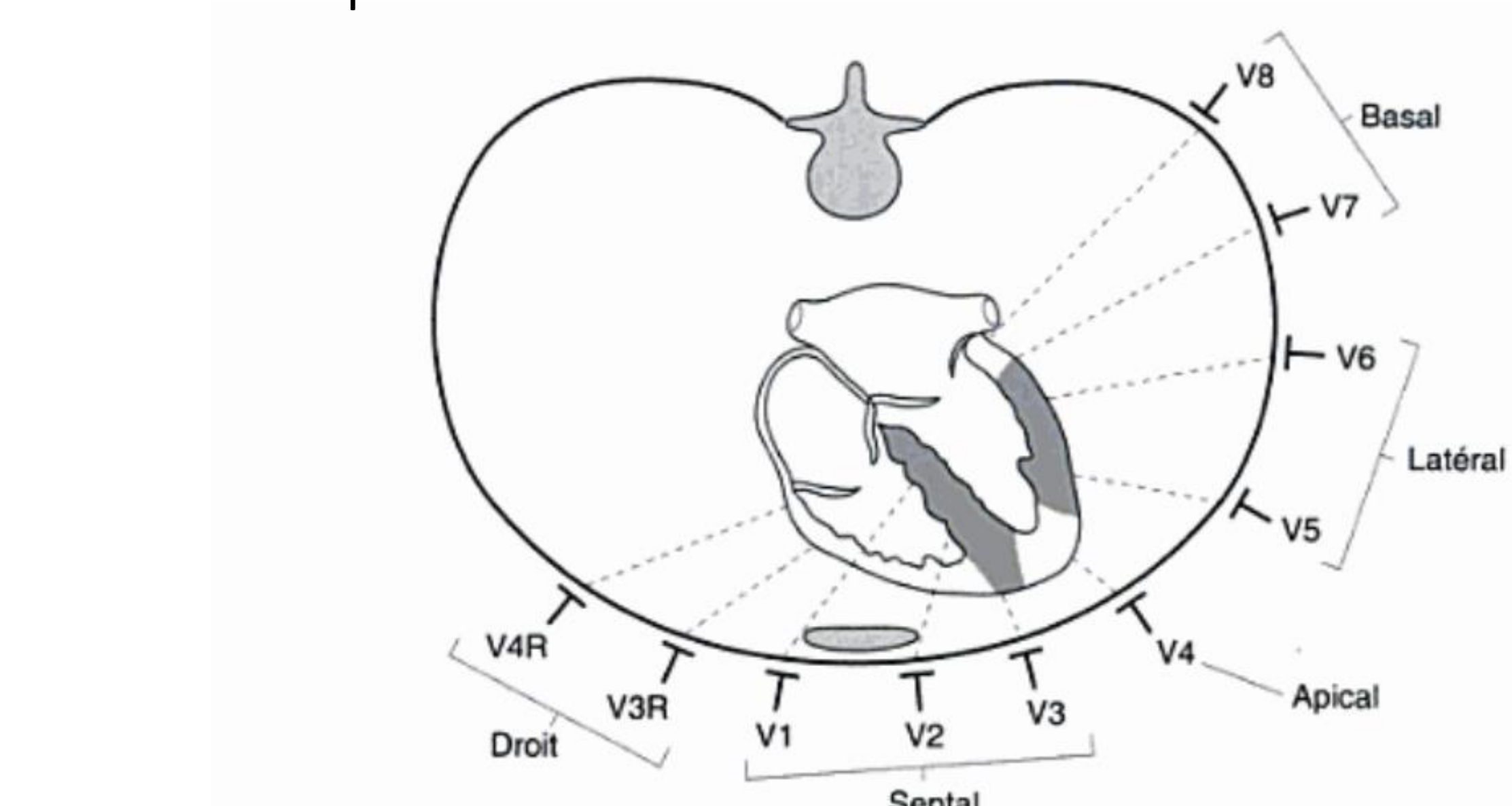



A
Normal heart rhythm recording (EKG)
R
P
T
Q
S
B
Nurse monitoring EKG procedure
Heart rhythm recorded on a machine to produce an electrocardiogram (EKG)
Electrodes connected to EKG machine
Electrodes attached to chest, upper arms and legs

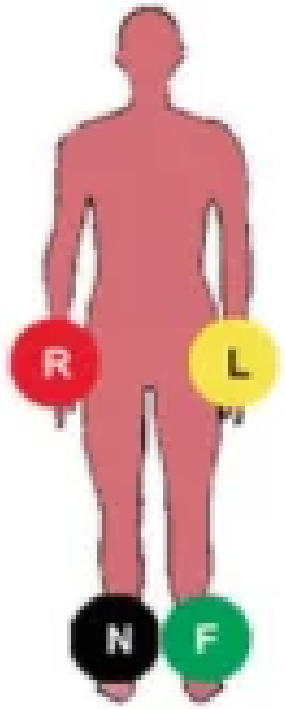


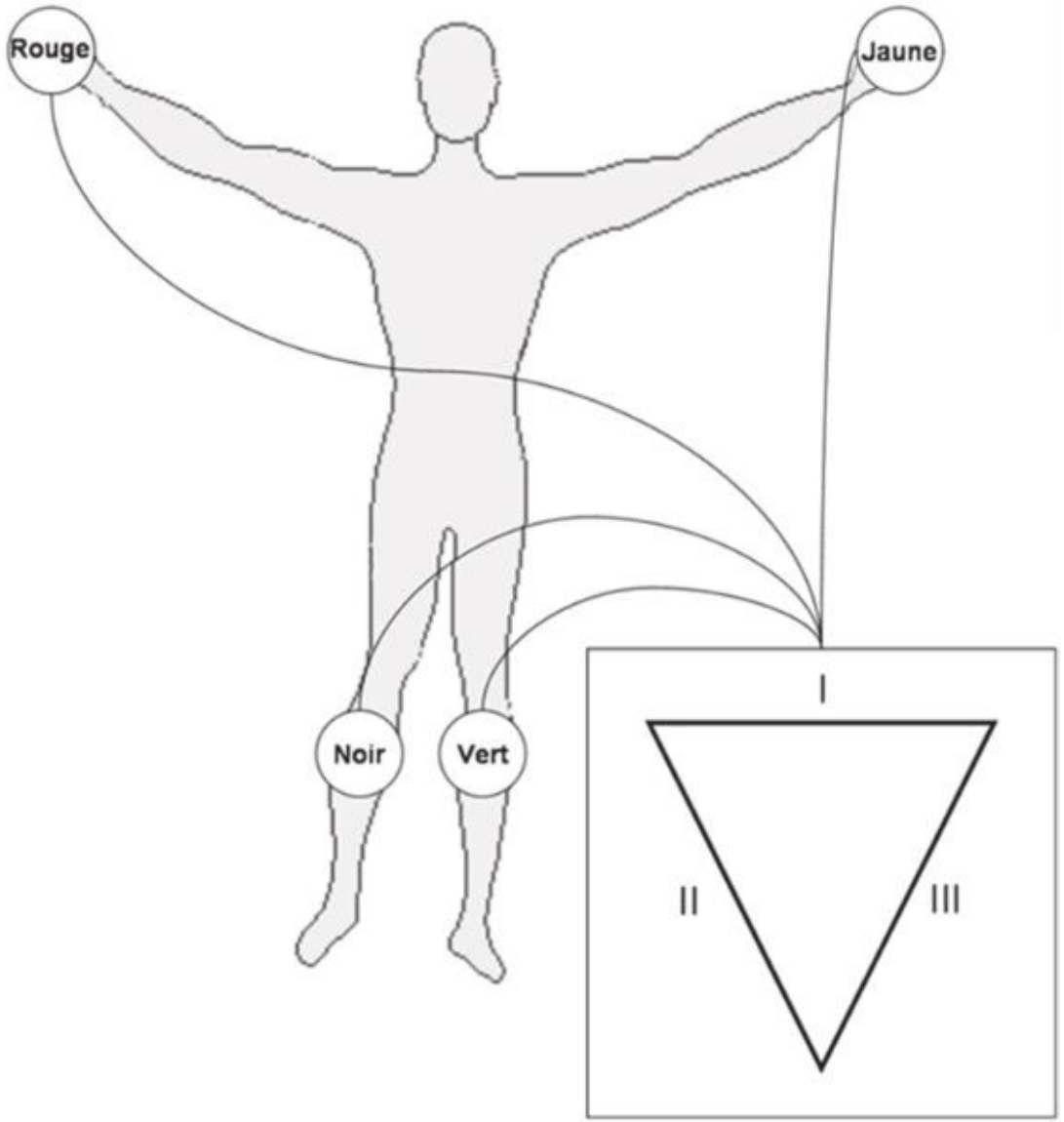
Placement électrodes précordiales

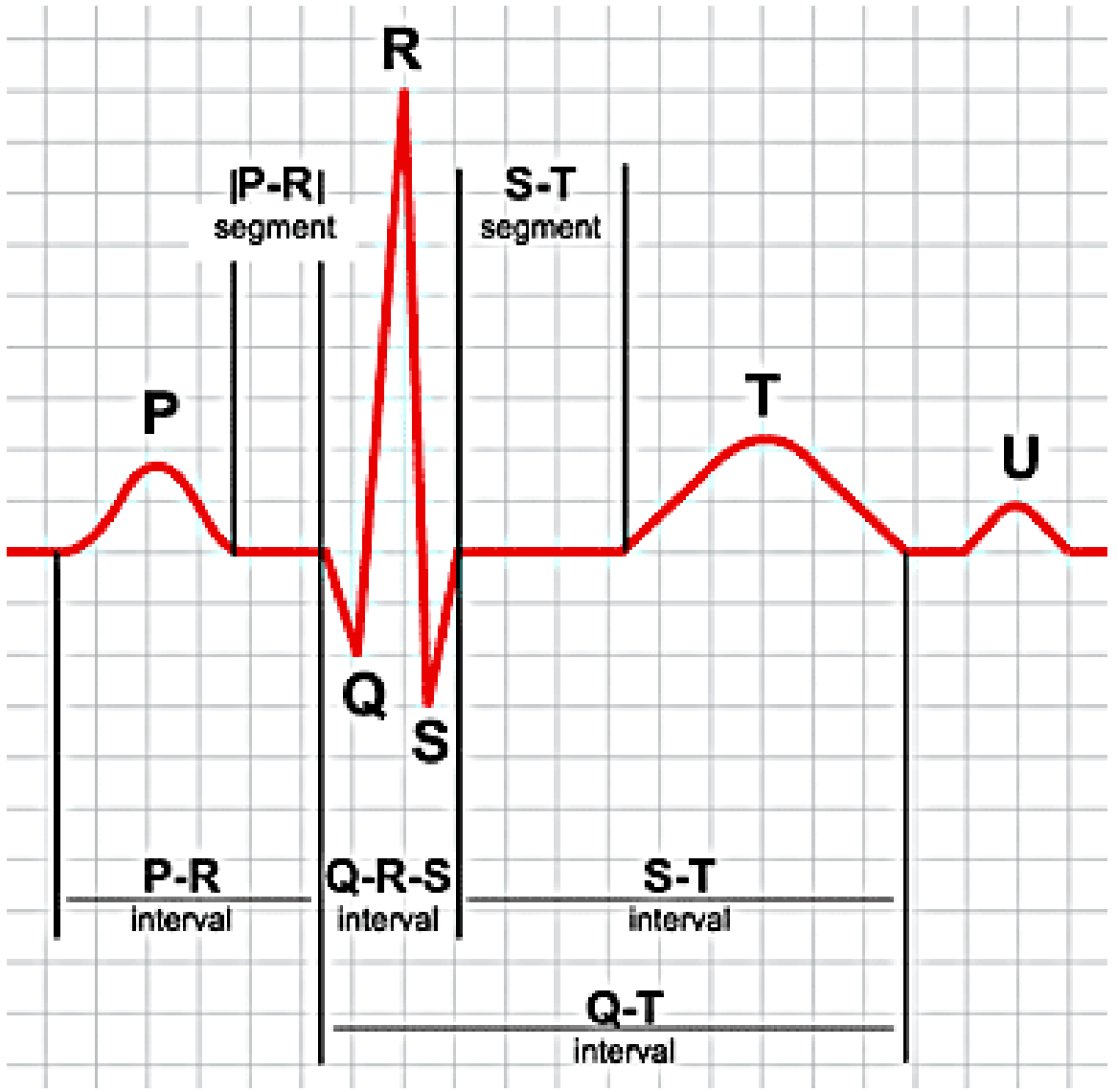
Dérivations précordiales

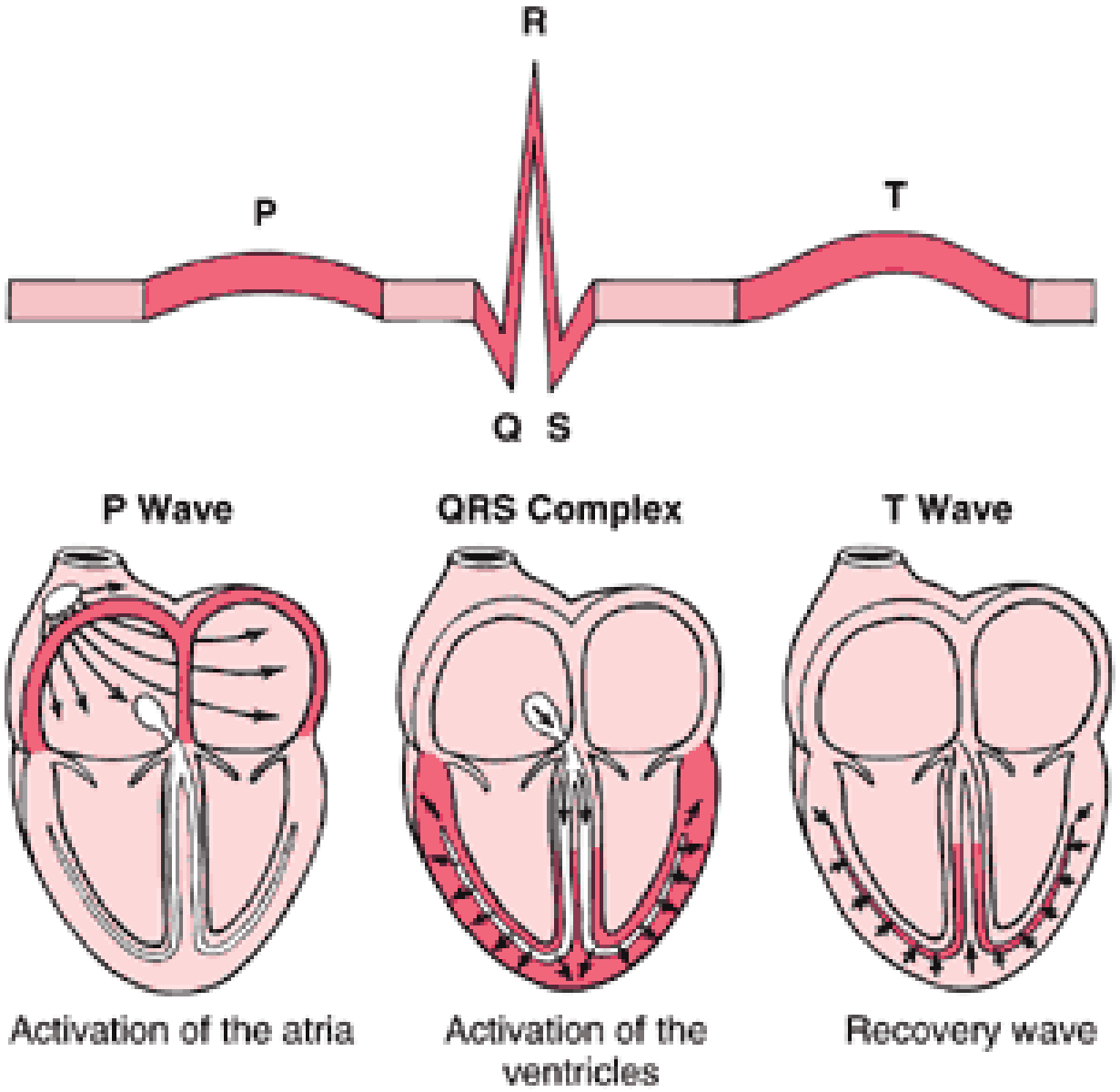


Dérivations frontales

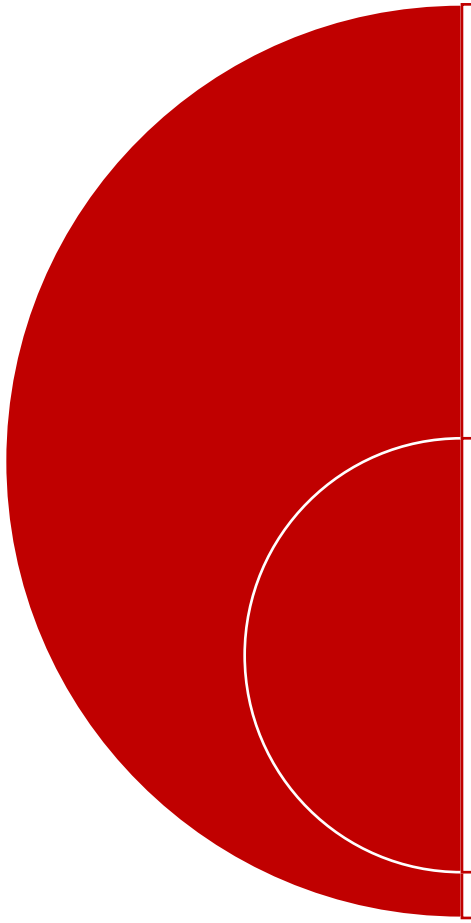





R
|P-R|
segment
S-T
segment
P
T
U
Q
S
P-R
interval
Q-R-S
interval
S-T
interval
Q-T
interval


R
P
T
Q S
P Wave
QRS Complex
T Wave
Activation of the atria
Activation of the ventricles
Recovery wave

indications



Tous les signes fonctionnel cardiaques :

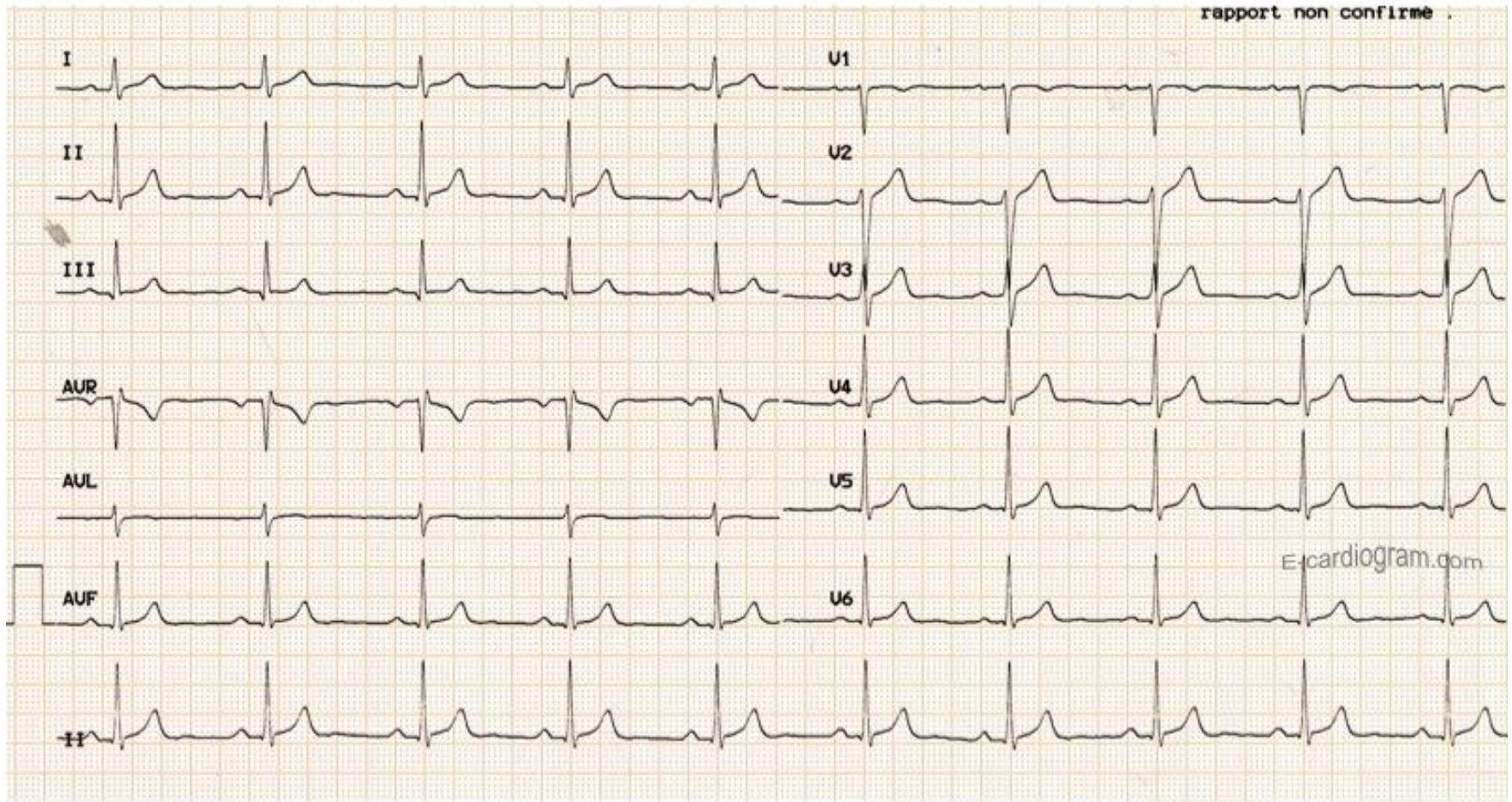
- Douleur thoracique
- Dyspnée
- Palpitations
- Syncope et lipothymie

indications

- tachycardie, bradycardie, pouls irrégulier
- souffle cardiaque
- Toute maladie cardiaque (Dg et suivi)
- insuffisance cardiaque
- HTA
- AOMI
- Diabétiques
- Bilan pré-opératoire
- Certificat d’aptitude sportive
- Troubles ioniques
- Intoxication médicamenteuses

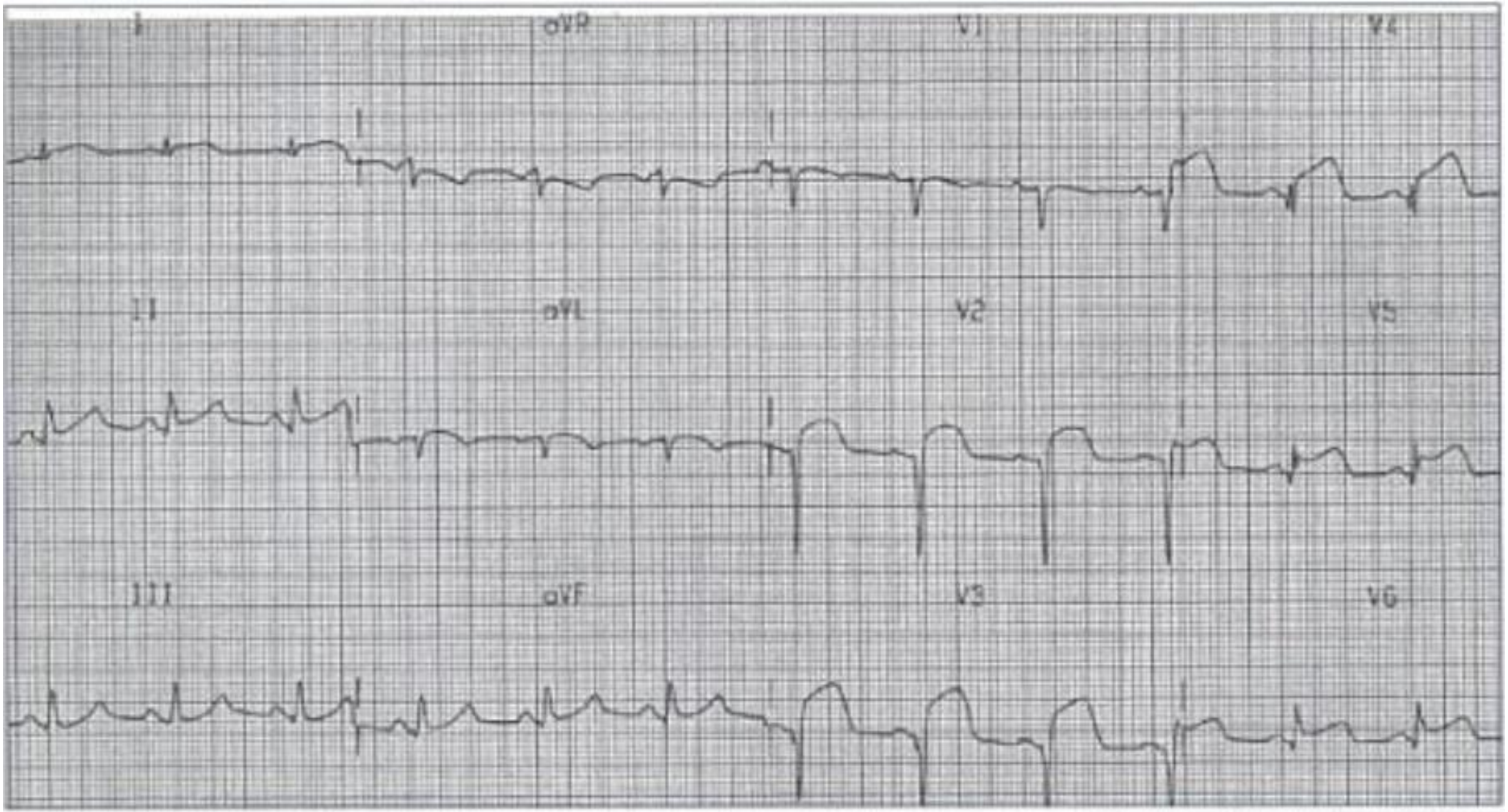
ECG normal

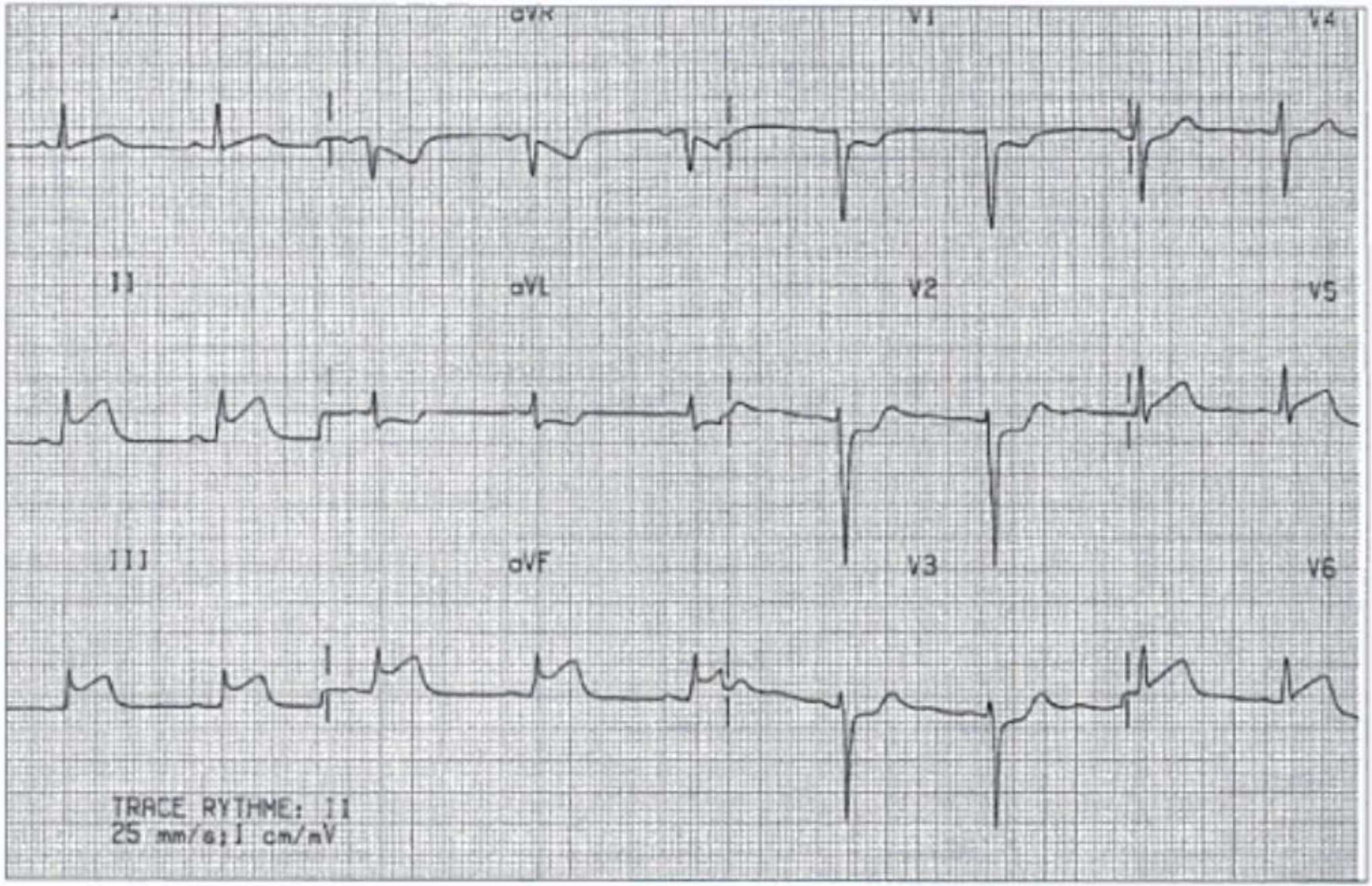
(homme 35 ans)

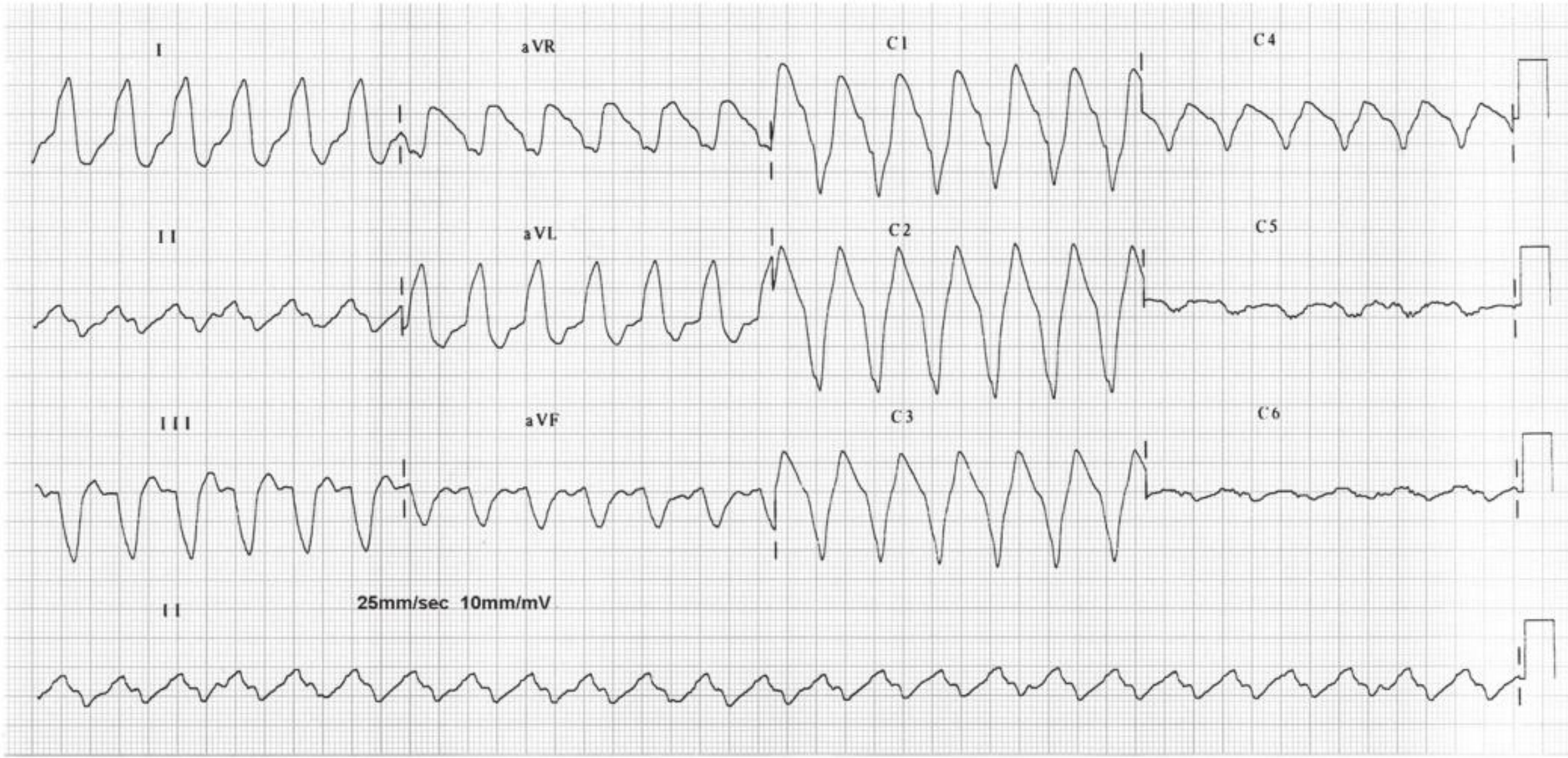


Normal Sinus Rhythm

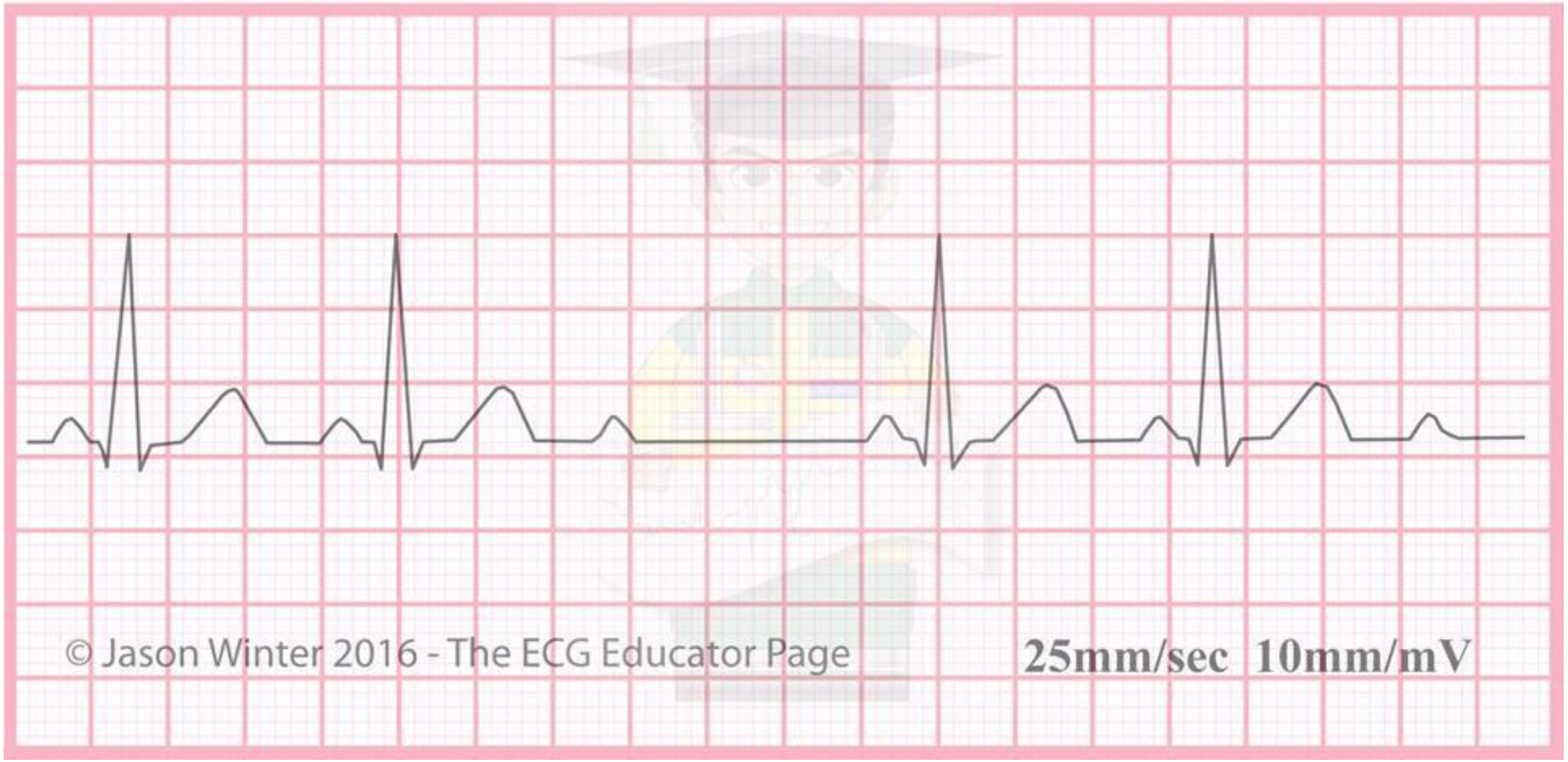



I
aVR
V1
V4
II
aVL
V2
V5
III
aVF
V3
V6


I
aVR
V1
V4
II
aVL
V2
V5
III
aVF
V3
V6
TRACE RYTHME: II
25 mm/s; 1 cm/mV


I
aVR
C1
C4
II
aVL
C2
C5
III
aVF
C3
C6
II
25mm/sec 10mm/mV

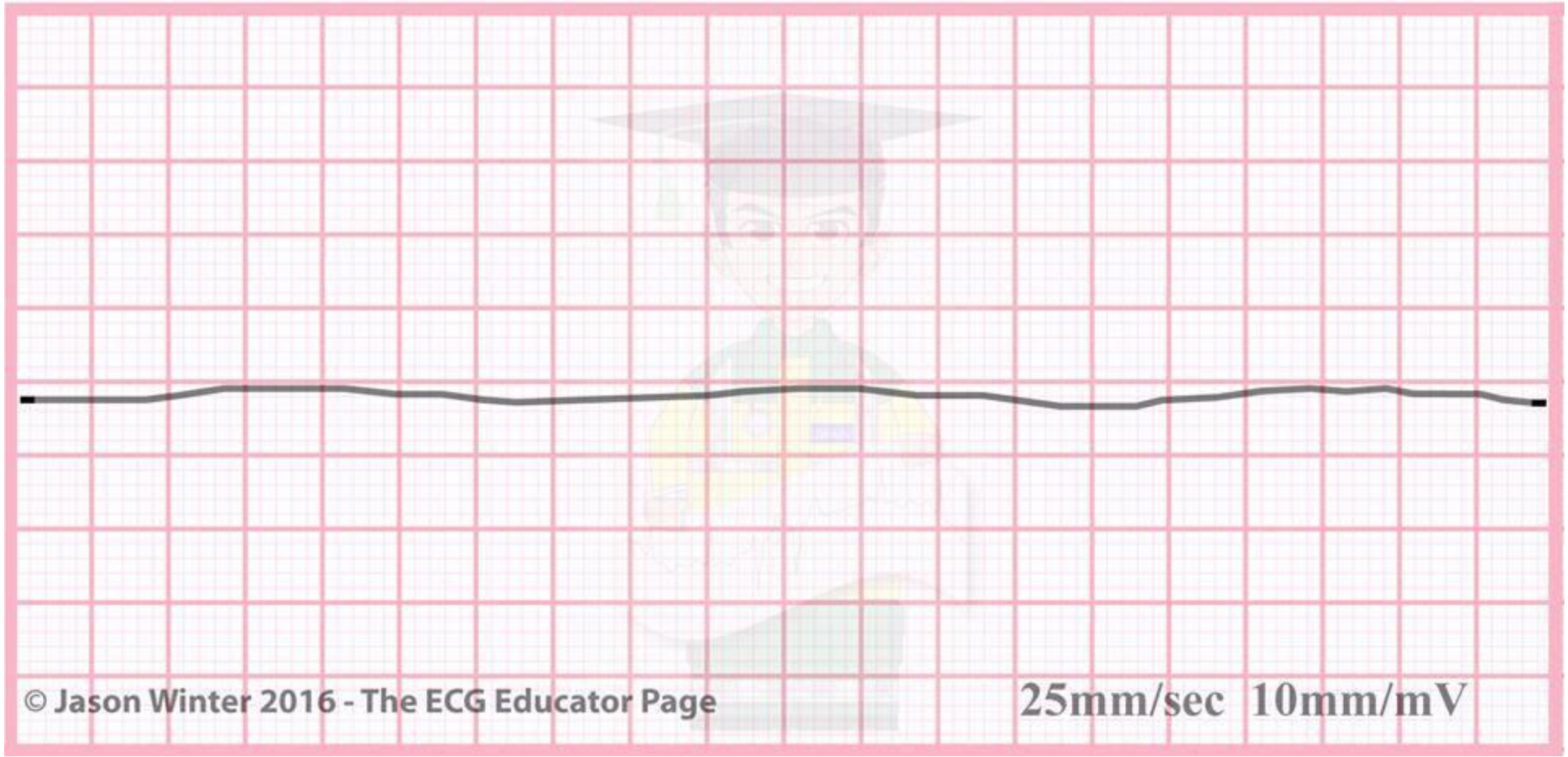
Second Degree AV Block - Mobitz Type 2



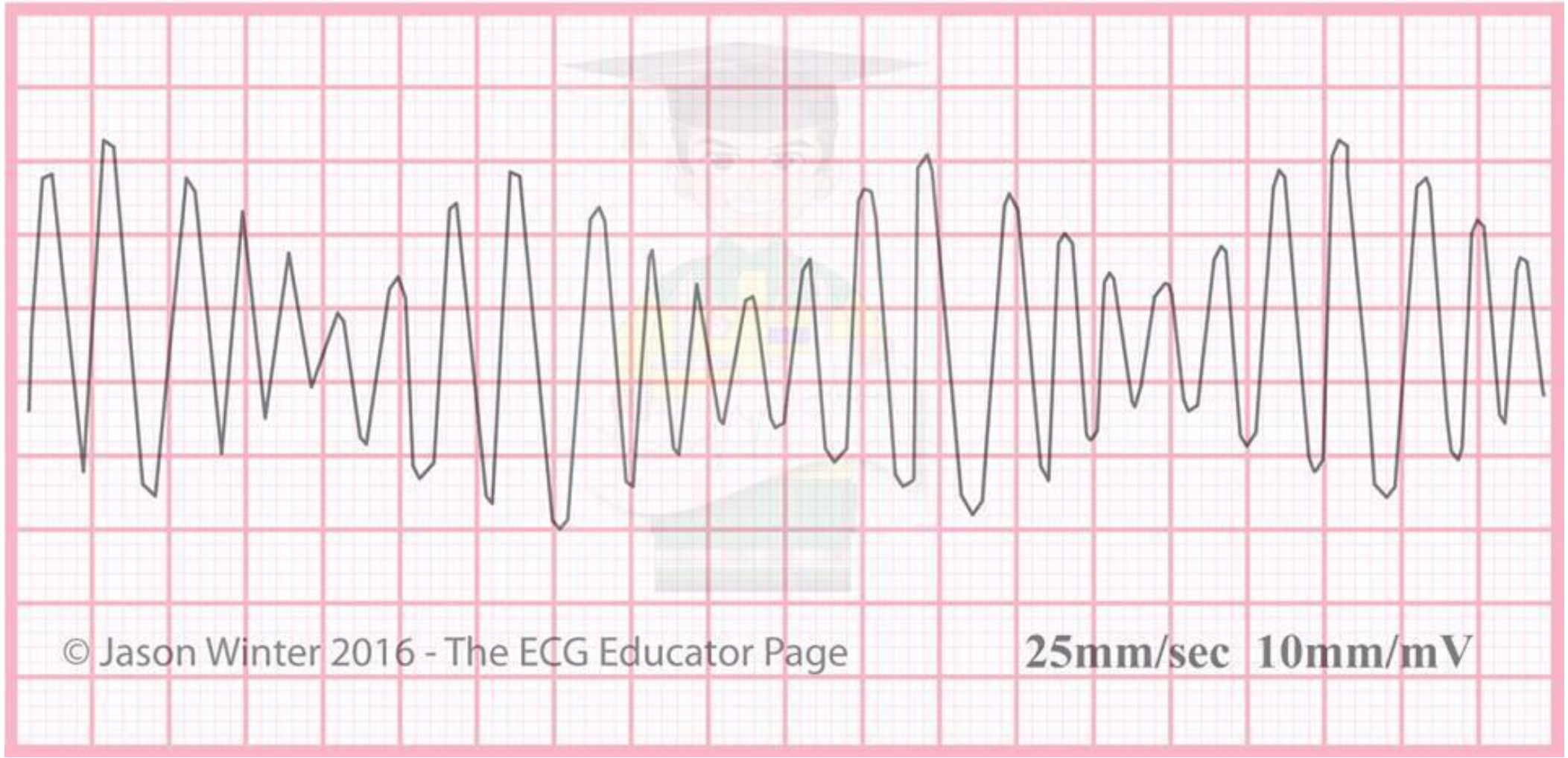
3rd Degree AV Block



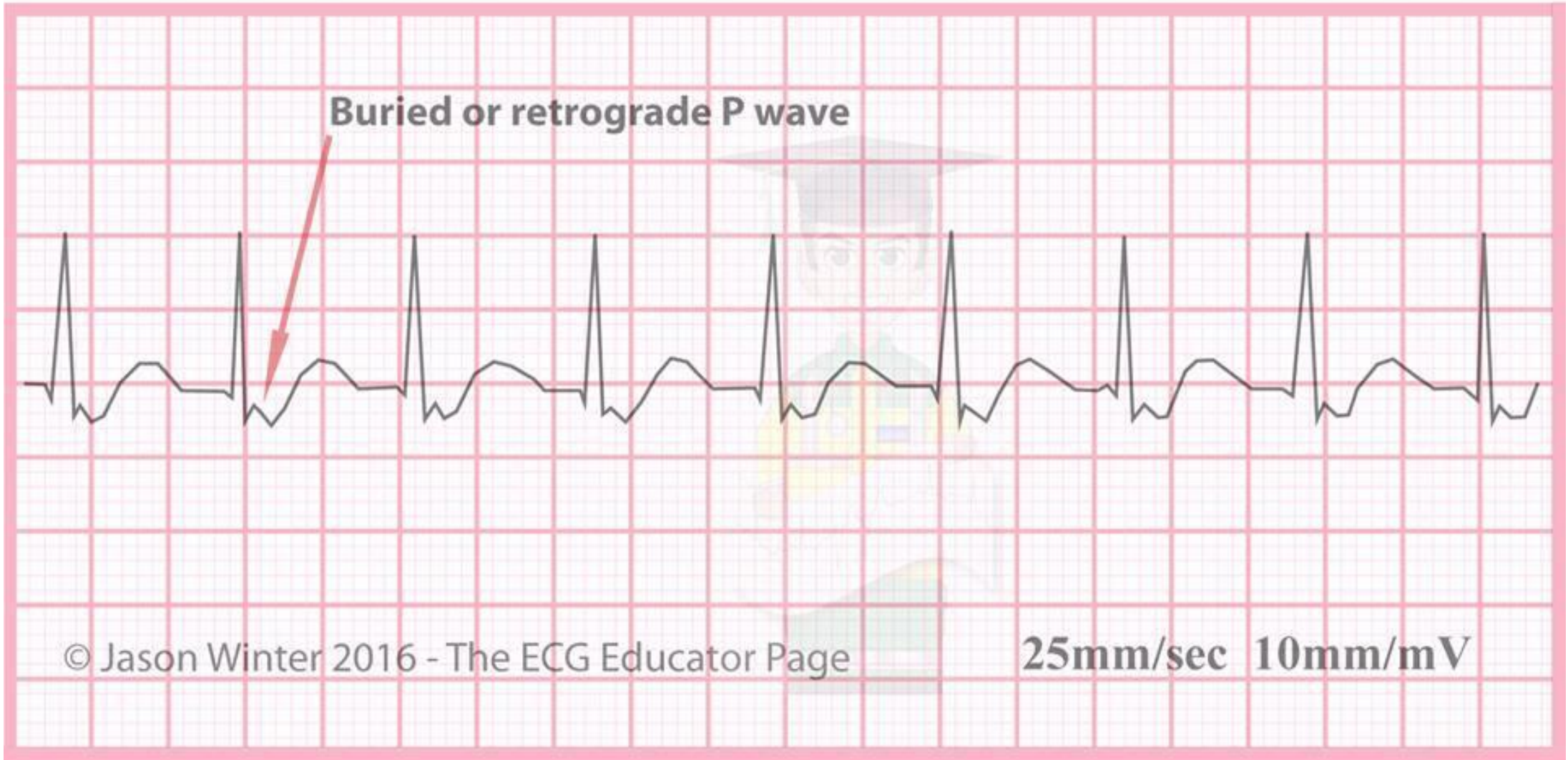
Asystole



Polymorphic Ventricular Tachycardia



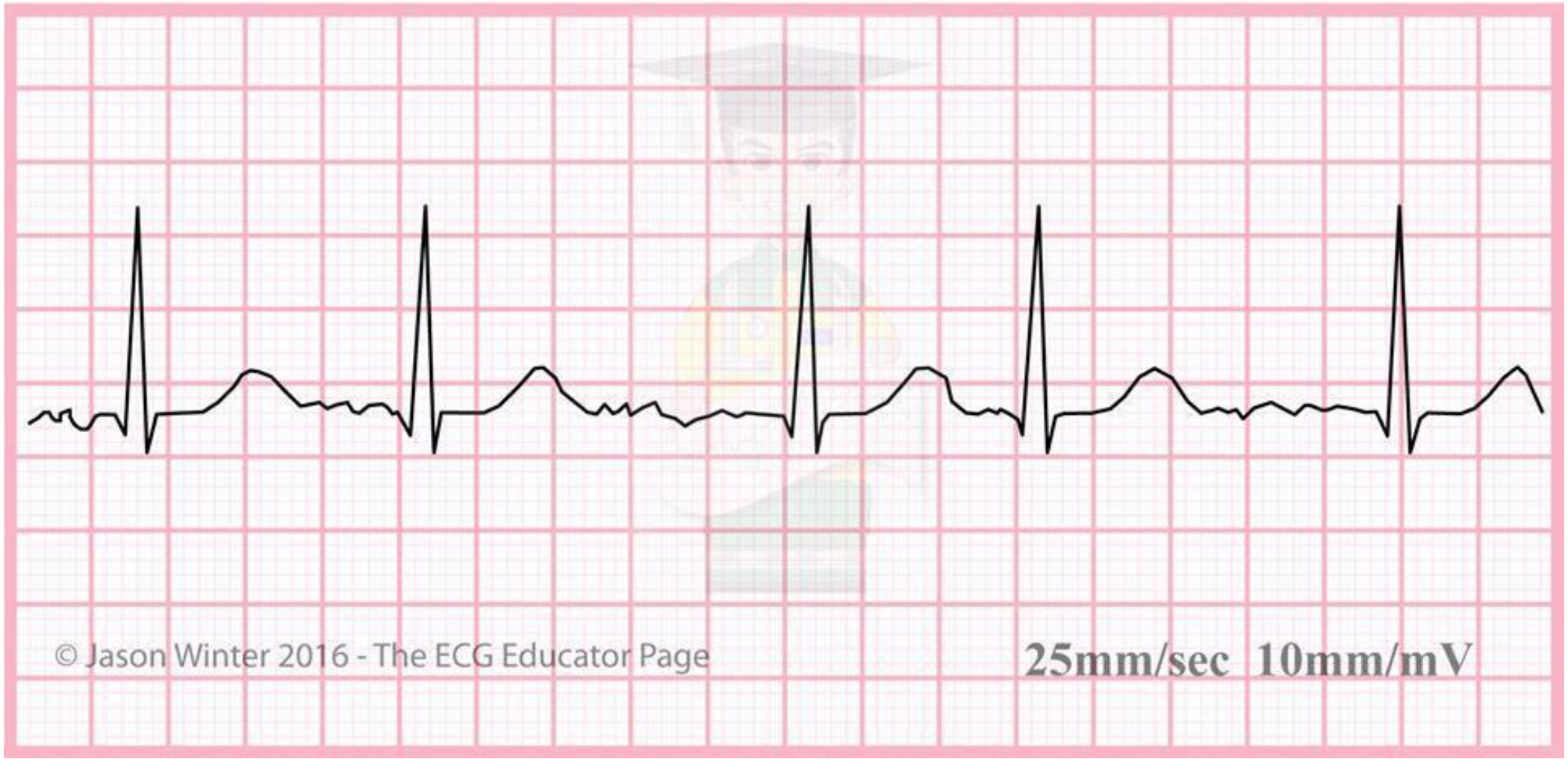
Junctional Tachycardia



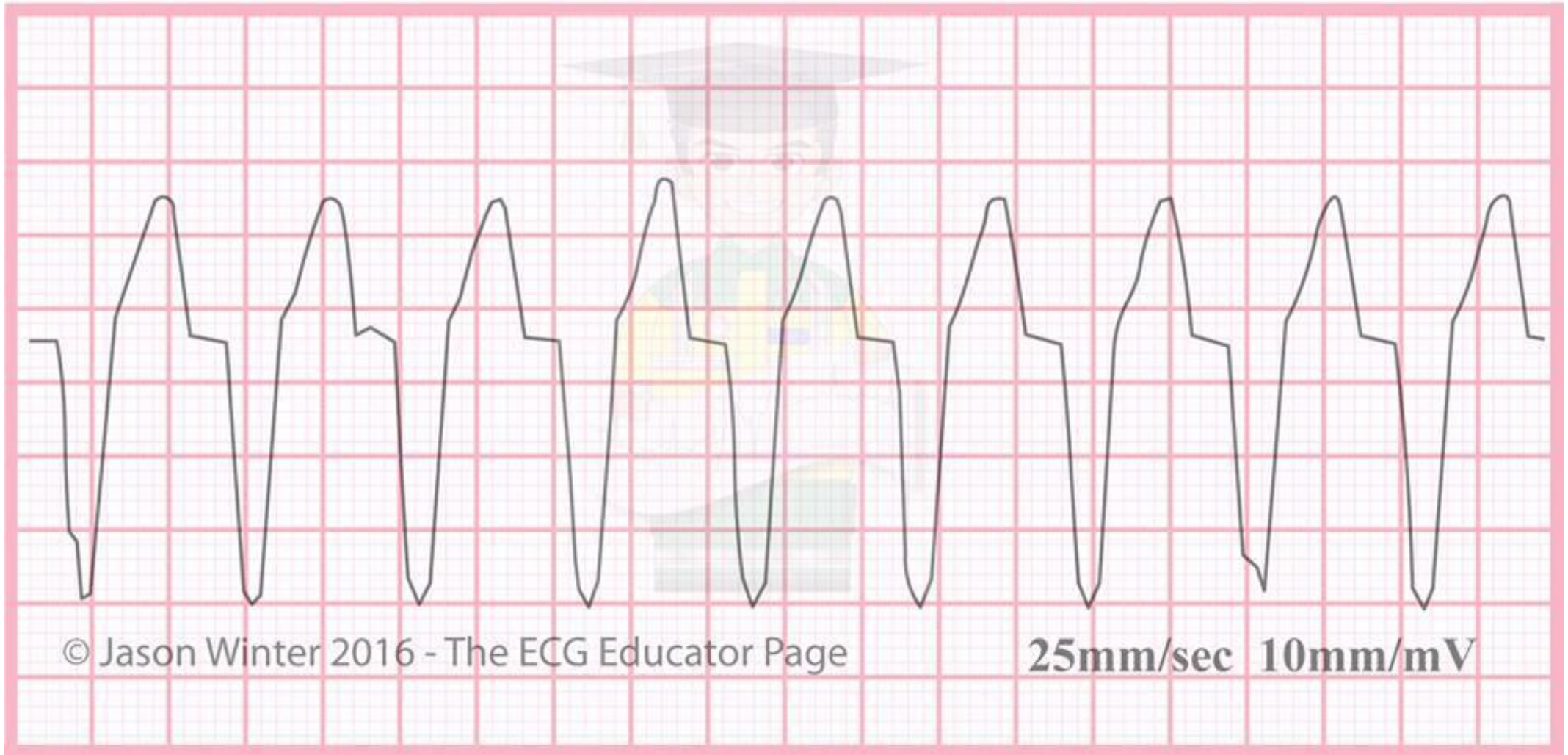
Sinus Pause



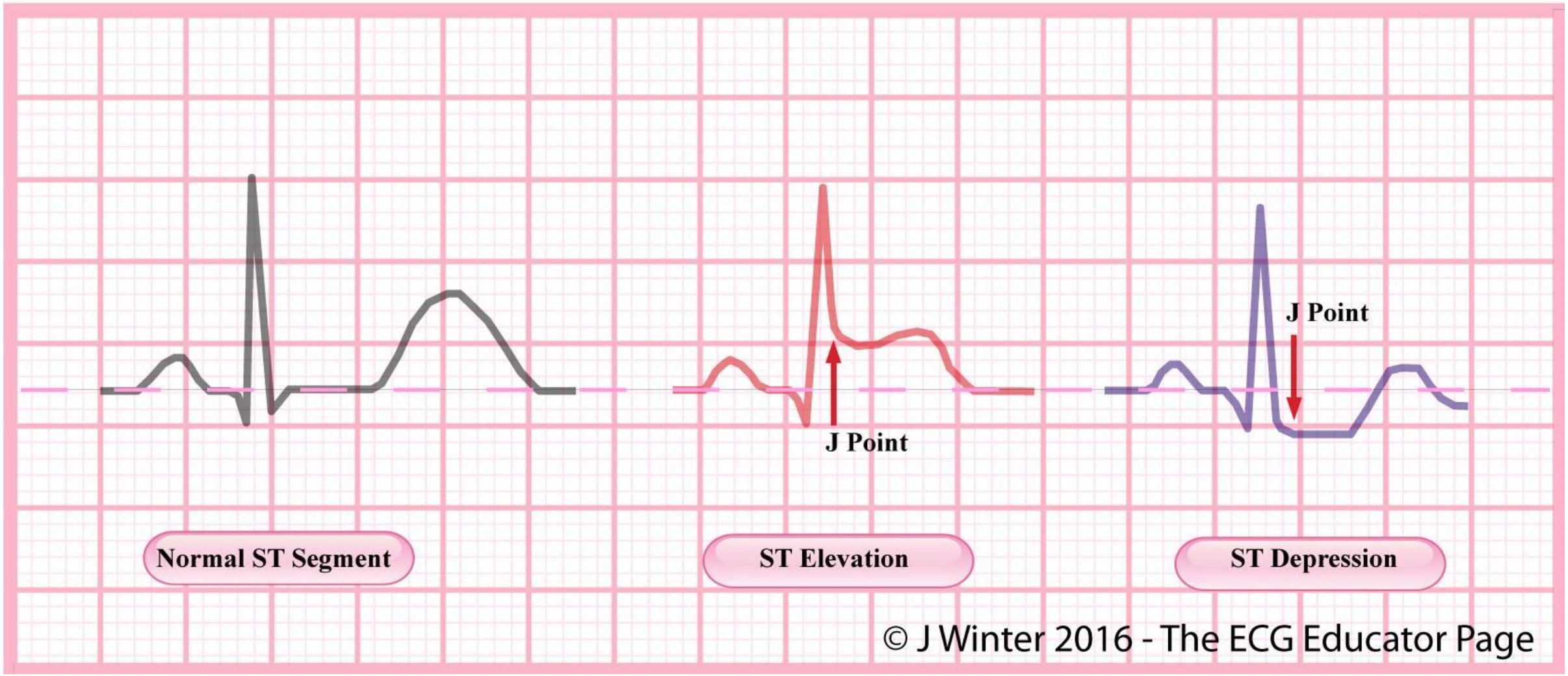
Atrial Fibrillation



Ventricular Tachycardia

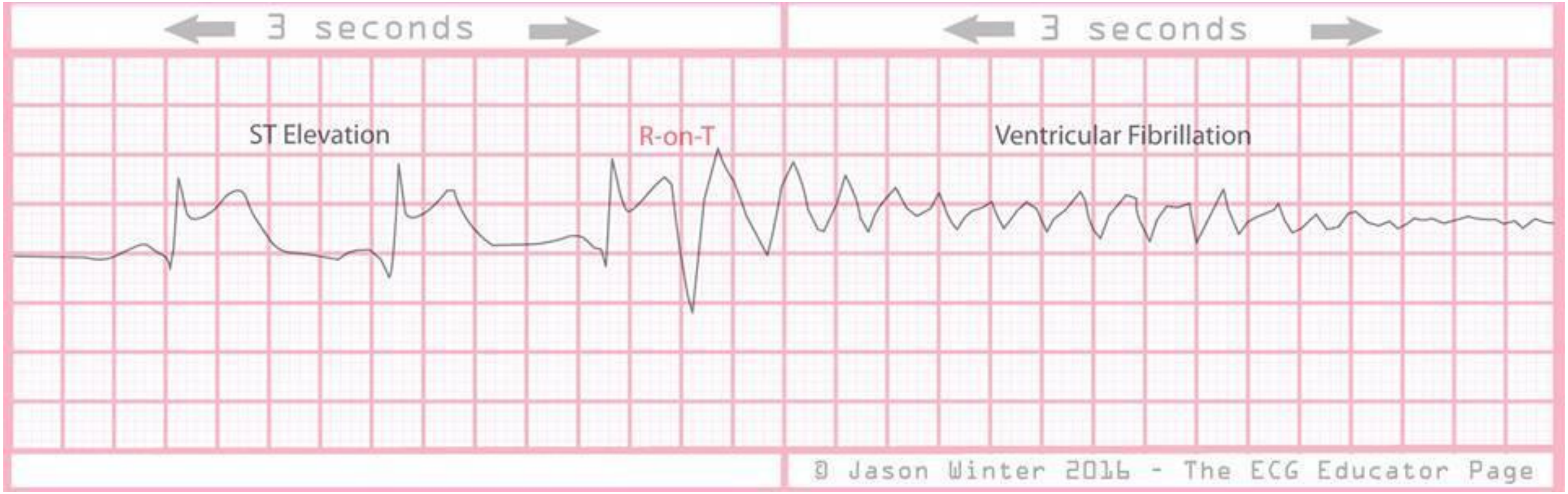


ST Segment Changes

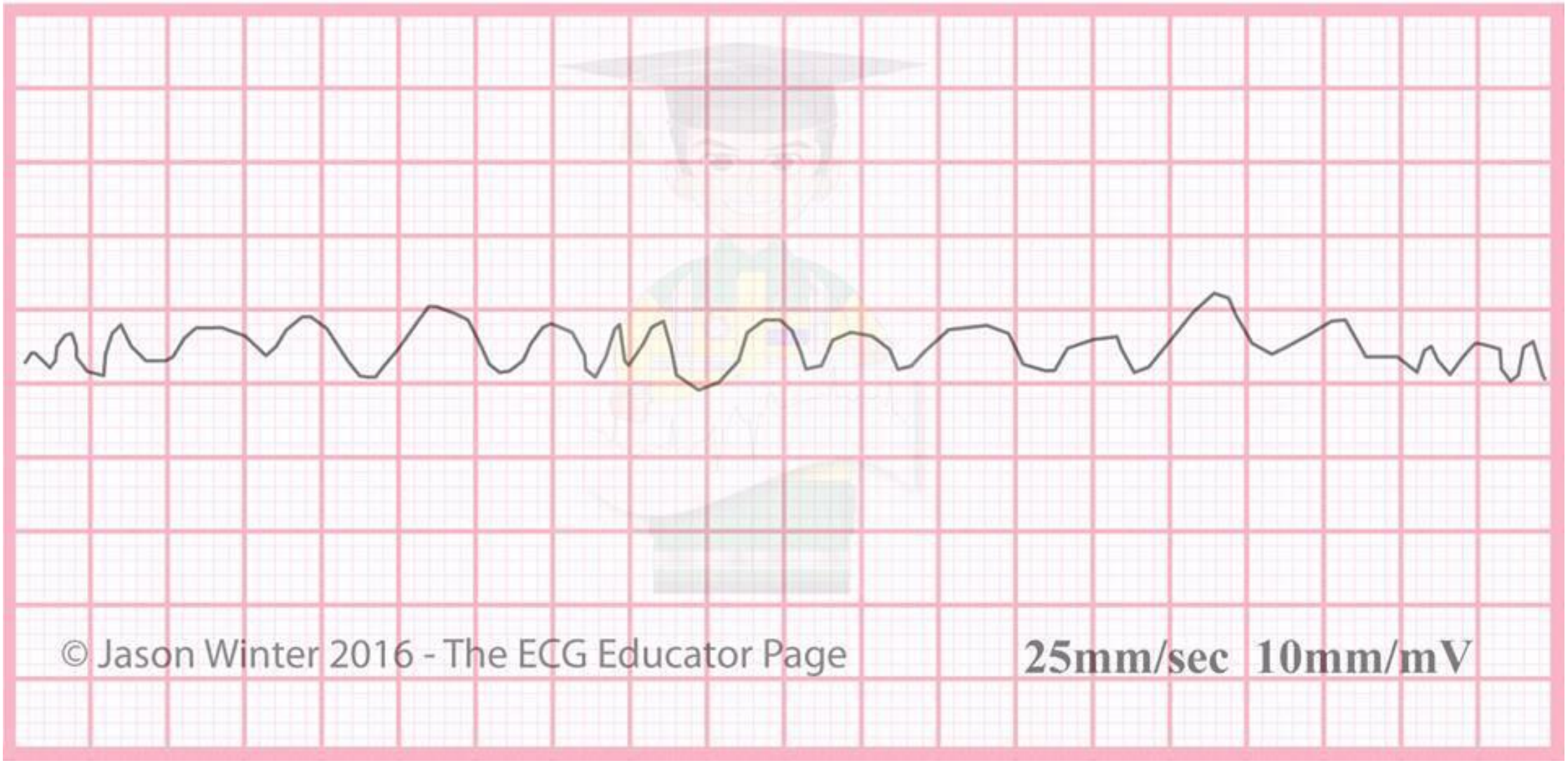


Atrial Flutter




3 seconds
3 seconds
ST Elevation
R-on-T
Ventricular Fibrillation
© Jason Winter 2016 - The ECG Educator Page

Ventricular Fibrillation



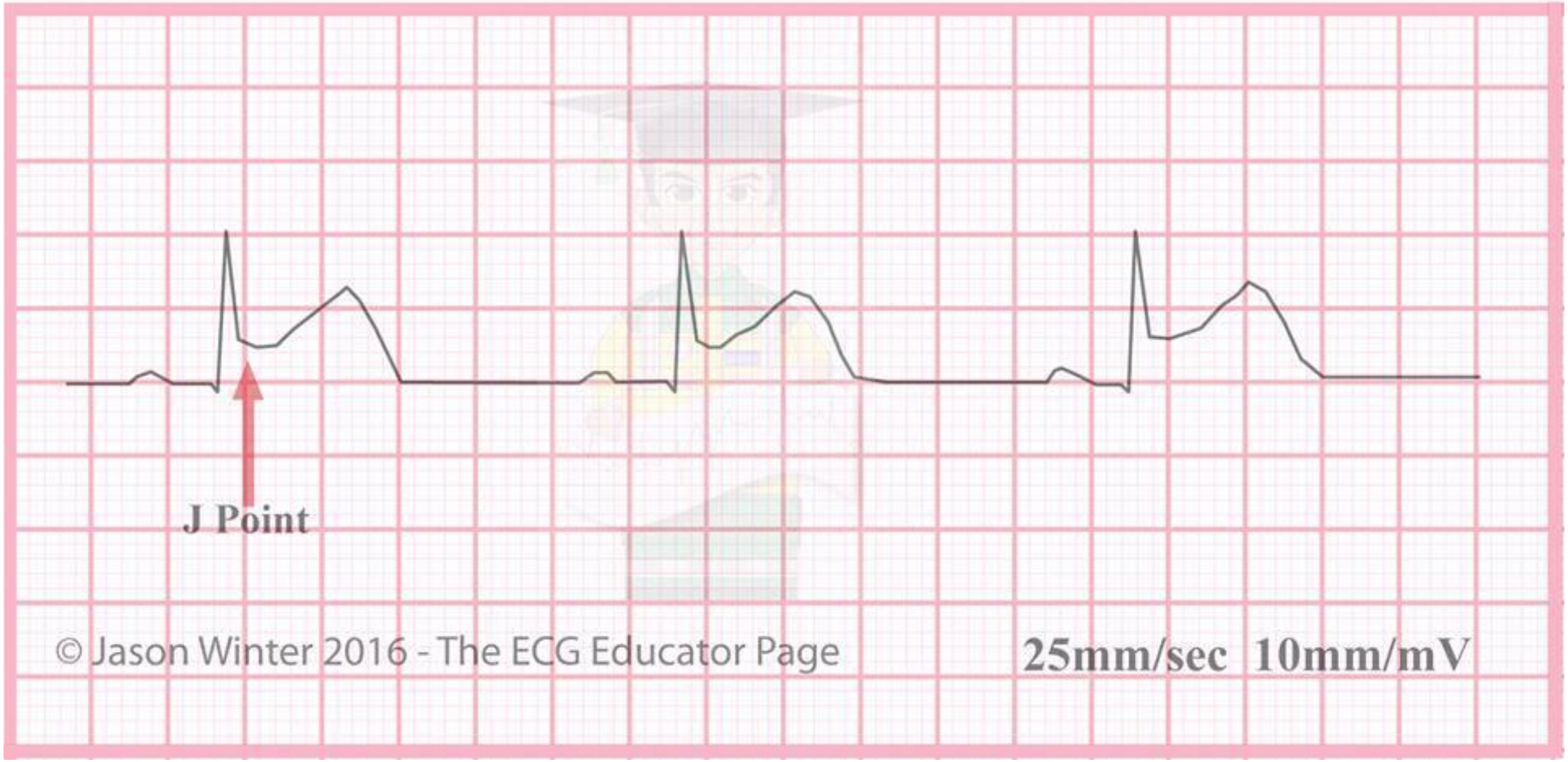
NSR with isolated PVC



Sinus Bradycardia



ST Segment Elevation



Merci pour votre attention